BEHAVIOR AND MANAGEMENT OF DOGS

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Objects of keeping dogs:

- Housed hold animal.
- Guarding (watch) dog in the farms; factories; police and army.
- Detection of estrous in cows.

Housing:

- Should be provided with a box or basket
- Placed in a corner of a kitchen where there is warmth and freedom from draughts.
- Bedding should be provided which may be an old blanket or a piece of carpet.
- When a number of dogs are kept they are best accommodated in sheds or buildings apart from the house.

Feeding:

- The amount and composition of diet is related to: size, age, amount of exercise and mode of life; used for breeding or not and the breed.
- Daily feed intake of adult about 3 to 5% of their body weights.
- Puppies require a daily feed intake of about 7% of body weight.
- The number of feeding for the adult divided into 2 or 3 times daily at regular intervals or may be given one large meal

- The puppies from birth to two month of age should be given 6 meals/day, then from 2-4 months fed 4 meals/day and after 4 months given 2 meals/day as adult.
- The protein % in the diet should be 17-20% for adult which are in the form of minced meat
- 25-33% for young and bitch during lactation which may be in the form of milk and yoghurt.
- While the diet should contain a fat % not exceeds 5% and water% not exceeds 80%.
- If the dog suffering from renal insufficiency give him meal low in protein % for not increasing level of urea and ureates.

- Row eggs are very nutritive for dogs.
- Sweats and fruits are not necessary for them.
- Raw fish as led to nervous manifestation and paralysis due to some enzymes.
- Give them large bones for chewing and sharping of teeth, also increase amount of saliva which help in digestion and a source of minerals
- Not give them bones of chicken and rabbit as splinted into sharp particles which injury the mouth and cause impaction and constipation for dogs.

- Should be supplied with vitamins in their diet within the recommended requirement
- Its deficiency or excess led to problems for dogs
- As Vit. D. deficiency led to bone dystrophy
- While it`s excess led to calcification of soft tissues
- Vit.E. deficiency cause young born dead or weak or affected by cardiac insufficiency and also muscular dystrophy.
- Deficiency of Vit.B1 cause paralysis or neurosis or convulsions
- Niacin deficiency led to canine pellagra (Black tongue)
- Vit.C. deficiency led to scurvy.

- We give them vermifuge by a dose monthly to protect them from internal parasites
- Clean and wash their coat to prevent manage and dermatophytes.

Drinking:

- Dogs weighing about 20 kg ingested about one liter of water per day.
- 0.5 liter by drinking and the rest in food.
- They drink about 9 times per day of 60 ml of water every time.
- Variations according to many factors, like food, weather ..... etc.

Exercise:

- Dog is taken for a short walk once or twice daily. Exercise on grass is preferable than on roads, which cause much wear and tear of the pads of the feet.

Aggression types:

1.Social aggression:

- Submissive dog show ears back, tail lowered between legs, lowered body posture with increasing fear and in extreme passive submission the dog will lie down on its side, lift its hind leg and urinate.
- Dominant dog have erect tail and ears in T-position to submissive dog`s shoulder
- So submissive dog turn head away avoiding eye contact exposing jugular area to dominant dog and submissive dog remain stationary as running will lead to attack.

- Dominance is determined by size, temperament, sex as male is dominant over female although there is breed difference as female Shetland sheep dogs is dominant over males but not female fox terriers and breed. In puppies dominant hierarchy formed at 4 weeks of age.

2.Territorial aggression:

There are breed difference as German shepherd rank first in interspecies aggression although they rank sixth in population.

3.Fear-induced aggression:

Fear biting dogs will attack if its critical distance is invaded.

4.Maternal aggression:

In case of Pseudopregnancy under influence of high progesterone level bitch take inanimate object as puppy and guard it viciously

Rest and sleep:

- Dogs show short periods of activity interspersed with periods of rest both in free ranging and caged dogs
- In case of caged dogs 30 min-2hr active periods.
- The caged dogs spend half of their time sleeping (80% sws) and quarter of their time standing.
- Sleeping dogs lie in a characteristic posture with their hind legs tucked up and their heads turned caudolaterally with their eyes may be open or closed.

Breeding in bitch:

1.Puberty:

- Smaller the breed reach puberty earlier than the larger breeds.
- Puberty onset for all dogs ranges from 6 to 15 month with 7 to 10 month being the usual for the average dog.

2.Estrous cycle:

- The domestic dog is a seasonally mono-estrous breeder in spring and autumn from one to four cycles yearly may be seen, most usually two.
- The average estrous cycle length is 21 day.
- Divided into four stages (proestrous, menestrous, estrous and metaestrous).

The signs of estrous cycles as follows:

A. During proestrous: first 1-3 days

1.Bitch becomes attractive to male dogs, run with him and will sniff and lick the male body and genitalia.

2.Urination becomes more frequent as estrus approaches.

3.She does not accept mating.

4.The physical signs include swelling of vulva and appearance of mucous discharge from vulva.

B. During menestrous stage: 3-5 days

1.Swelling of vulva with blood stained mucous discharge due to rupture of engorged capillaries of vagina.

2.She does not accept mating.

C. During true estrous :6-18 day

1.Stand quiet and allow male to investigate and mate her and sometimes she present her hind quarter toward him or mount him to attract him.

2.Mucous discharge becomes transparent and watery with frequent urination.

3.Swelling of vulva and she accept male mounting at day 8-9 and day 17-18.

D. During metaestrous stage :18-21 day

1.She does not accept mating.

2.Disappearance of vulvar swelling and mucous discharges.

Breeding in dog:

1. **Puberty:** in 5 week old male puppies.
2. **Courtship behavior:**

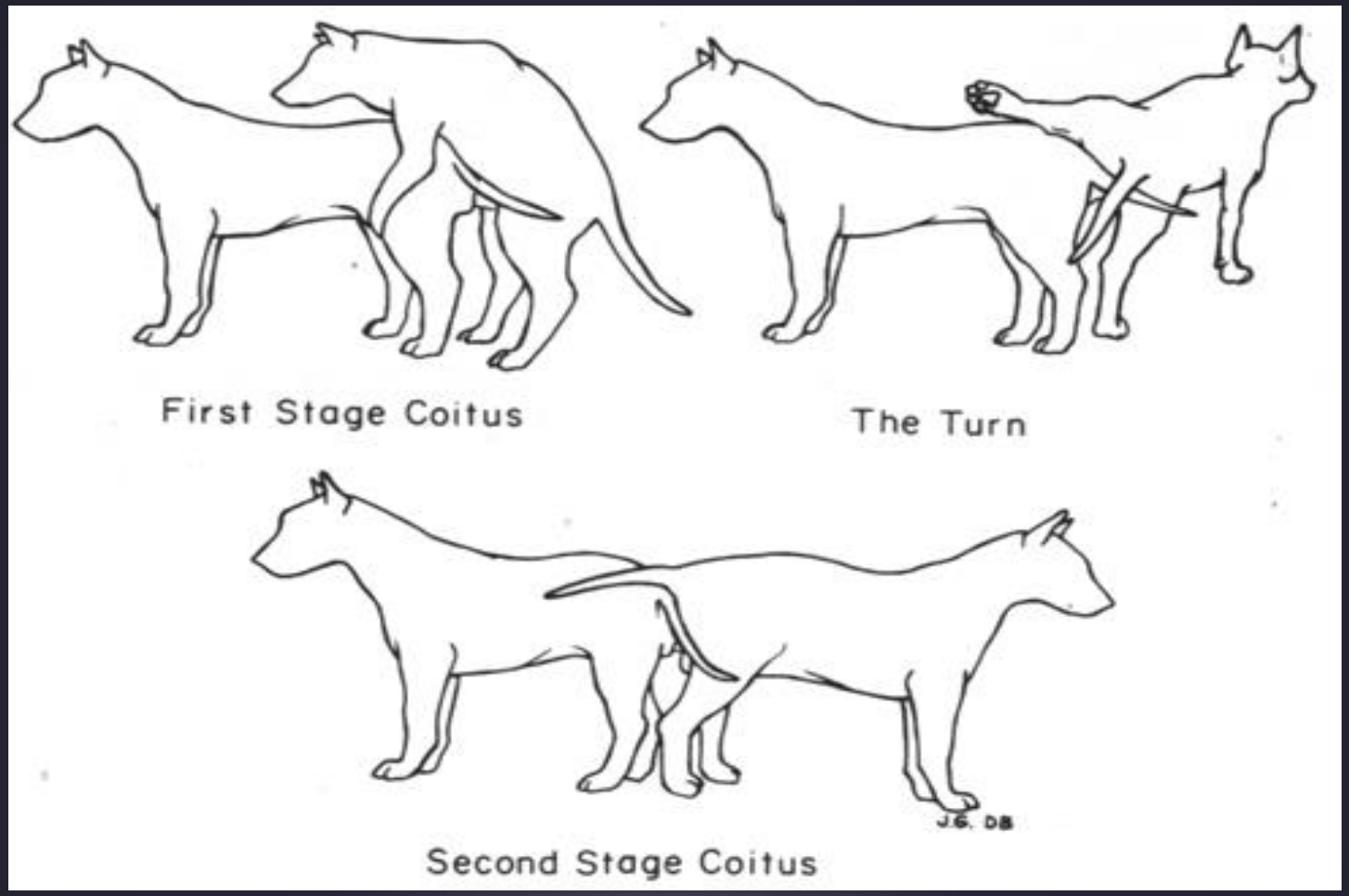
- Series of reciprocal interactions between male and females before mating.
- Male dogs are attracted to urine and vaginal secretions of the estrous bitch but he is more attracted toward urine.
- Due to the presence of pheromones (methyl-hydroxy-benzoate), which are considered the "releaser" of sexual behaviour in the male.

The courtship behavior includes:

- The male sniffs the female's head and vulva; he may lick her ears.
- Don't show the classic Flehmen response of ungulates, it is possible that the **tonguing** response (bringing the tongue between the incisors) seen during this olfactory investigation.
- The male mounts in response to female immobility, he grasp her with his foreleg just cranial to her pelvis.
- He thrusts with his pelvis and when intermission has been achieved, the rate of thrusting increases.

- Engorgement of the bulbs glands with blood and contraction of the vestibular vaginal muscles following intromission result in the **copulatory lock** or tie.
- Prevent penis with drawl so prevent descend of semen, a phenomenon most closely associated with canine, but not restricted to them.
- The lock may last 10 to 30 minutes (mean = 24 minutes)
- The copulatory lock in other mammals seems to be a weak tie of 40 to 45 seconds.
- The male will dismount and turn around so that male and female are facing opposite direction while ejaculation occurs.

- After that the bulbs decreases in size and the pair separates.
- Young female was bred by older male while the older female was bred by young male
- A bitch may display dominance over male by not allowing male to stand over her or to approach from behind.
- Mounting is a sign of dominance,
- Submissive animal will stand for a more dominant one
- Female preference for certain males depending on the form of sexual display or the temperament of male.
- Dominant males were not necessary chosen as preferred mates



Coital position of the dog.

Gestation: The average period is **sixty-three days**.

Signs of pregnancy:

1.Eat small frequent meals (4-5 times/day) as abdominal pressure increases.
2.The activity decrease during the late pregnancy.
3.She grunts during sleeping due to pressure of fetus especially if she carries a larger number of litters.
4.Increase the size of abdomen and mammary gland

Management during pregnancy:

1.The bitch during pregnancy should be treated normally, jumping up or down and severe exercise should be avoided.

2. Food should be good quality and not given in large amounts. Mineral and vitamin-rich supplements for the last half of pregnancy should be provided.
3. Constipation should be prevented and drastic purgative may induce abortion.
4. From the end of the 8th week onwards, the abdomen and the hind quarters should be washed and carefully dried each day. Subsequently, the teats should be manipulated and rubbed with lanolin.
5. For the last two weeks, only gentle exercise and smaller but more frequent feeds should be given. Whelping takes place in the breeding kennel.

Pseudopregnancy:

1. Non-pregnant bitch during the luteal phase of the canine reproductive cycle
2. Some dogs show only slight enlargement of the mammary glands while others go through pseudoparturation at 49 days and actually lactate.
3. The pseudopregnant bitch is less active mimicking the slowing down of the truly pregnant animal and she may make a nest.
4. Estrogen therapy will relieve the signs of Pseudopregnancy but ovariohysterectomy should be recommended.

Parturition (Whelping):

a. Signs of approaching parturition:

1. Nest building: a nest box and nesting material such as strips of paper should be provided, so the bitch will make a nest in the last week of pregnancy.
2. Restlessness, inappetance and a drop in body temperature are the cardinal signs of impending parturition.

b. Onset of parturition:

1. Once labor begins the bitch lies in lateral recumbancy.
2. As labor progresses her hind leg will twitch due to uterine contraction
3. She will shiver and several abdominal contractions then occur rupture of fetal membranes and expulsion of pup.

4. The fetal fluids are attractive to the bitch. She will lick herself, any soiled bedding and the pup head, umbilical cord and premium which is the area’s most frequently licked.
5. Puppies are usually born at 30-minutes intervals, but delays of a few hours are not pathological.
6. If the bitch is disturbed; as mild as the entrance of another human observer in the middle of labor, it may stop, depending upon the temperament of the dog, 15 minutes to an hour are required for normal parturition to proceed.
7. If the placenta has not been delivered she may extract it by pulling on the cord. The placenta is then eaten.
8. She is usually silent as labor progresses unless there is dystocia. Adult male dogs, if present, often whine.

Maternal behavior:

a. Grooming:

1. The bitch cuts the umbilical cord with her molars and then licks the stump. She may recut the umbilical cord if it is too long.
2. Once each pup are born she will quietly lie and allow to nurse. She helps to orient the pup to her by licking.
3. Bitches seldom carry their puppies. Instead they lead them back to the nest by licking the pup's head.
4. Licking serves three functions in puppies. The licking **stimulates the pups to eat,** and when it is directed at the anogenital area it **stimulates urination and defecation** that would not occur spontaneously. **The third function of licking is retrieval.**

5. The bitch keeps the nest area clean by consuming the urine and the feces of the puppies for at least the first three weeks of their lives.

b. Nursing:

1. The puppies move toward the dam by paddling and turn their heads from side to side, as they cannot lift their bodies from the floor yet.
2. Textural and olfactory cues may help puppies locate their mother and her mammary glands.
3. When the pup locates a nipple, it doesn't grasp it, but noses it from beneath. Unless it opens its mouth at the same time, the nipple moves. Their ability to locate nipples improve in the first two or three days of life.. The inguinal mammary gland is the preferred one and is the one approached by a single pup.

4. During the first few days the bitch spends most of her time nursing, reaches the peak at end of the first week which decreases with time.
5. During early lactation the bitch always initiate the nursing bout.
6. By 3 weeks the puppies initiate most of the nursing bouts as they are blind till 9-12 days of age in most breeds.
7. The litter will be weaned gradually at 4 to 6 weeks by the bitch, completed by 60 days.
8. Some bitches may regurgitate food for their pups during weaning. This helps to maintain adequate nutrition during the transition from a milk diet to a raw meat diet as their powers of mastication and digestion not matured enough.

Neonatal behavior:

- Puppies have two vocal signals: the whine and the grunt.
- The whine is emitted when the pup is cold, hungry, or separated from its litter or mother.
- Whining will stop immediately if the puppy's head and neck are covered with a warm towel or if it is again placed with its litter.
- The grunt is a pleasure communication that occurs after warmth or reunion with litters or dam.

Post-parturient care and management:

a. Bitch:

1.Washing the hind quarters of bitch after parturition with warm water to protect her from flies.

2. Give her a drink of water, milk or gruel which made by adding 3 or 4 tables spoonful's of meal with 1 pint of cold milk or cold water allowing to stand for 15-20 minutes, then starting boiling for 15 minutes and add salt or sugar.

3.Richer feed is gradually increased to bitch after parturition, so that a week or 10 days later she is onto full feeding.

b. Puppies:

1.The pups should be put in a basket with an old blanket under which is a hot-water bottle very slightly above body temperature.

2.At about 3-4 weeks give them milk replacer or yoghurt or minced meat in addition to regurgitation of food by dam.
3.Gradually, the food increased to the pups and weaning usually occurs at four to six or seven weeks.

4.After weaning, pups need 4 meals daily, and must get adequate vitamins and minerals if healthy growth is to be maintained.

5.Many pups suffer from worm infestation during the first **3** month of life. Vermifuge given to the bitch at the third week of pregnancy. Pups are warmed as a regular routine at 3-4 weeks, 8-9 weeks and again at 12 - 14 weeks old.

Castration, neutering, is the surgical procedure which removes an animal's testicles (or testes). The main reasons :

1. Prevent accidental mating and unwanted puppies.
2. Address certain medical conditions including testicular cancer and diseases of the prostate gland.
3. Or/and to address behavioural problems.

- As dogs grow during their adolescence (6-18 months) the levels of testosterone hormone increase so the secondary sex-related characteristics become more evident.
- These include mounting, leg lifting, urine marking and higher levels of aggression.
- The hypothesis is that removal of the testes reduces the production of testosterone and hence sex related behaviour.

Anomalous Behaviors

- **Destructiveness:**

Causes:

1. Absence of the owner from the home.
2. Lack of exercise.
3. Barrier frustration: caused by the presence of something very desirable on the other side of the barrier.
4. Breed incidence: ex. Beagles; German shepherds seem to be particularly prone to destructiveness.

Treatment:

1. Negative reinforcement: make the consequence of chewing on an object unpleasant e.g. Painting the chewed object with a hot soace of chili and cayenne pepper or by giving the dog bone that cannot be splintered because it transfer chewing behaviour to a naturally preferred object like bone. The most effective punishment is electric shock.
2. Companionship: relieving the loneliness relieves the dog get off its destructiveness.

3.It is important that the dog get enough exercise.
4. Progesterone therapy for a short period (10-14 days) to calm the dog.
5.Behavioral modification: instead of punishing for being destructive reward it for good behavior, this learns the dog the environment of it has been changed, non-destructive behavior will be rewarded and it will receive attention from the owner without being destructive. This method is best but only success by 50% as few owners have patience to continue training.

- **Barking:** barking is an even more difficult problem to treat than destructiveness because it affects not only the owner but anyone else it earshot. Some dogs tied up outside, bark because of discomfort. **Remedy** by shock collars that administer a shock whenever the animal barks have been used with some success but the severity of the shock is variable.
- **Worrying livestock:** A serious vice is sheep worrying. In addition to the sheep which may be attacked and killed or injured, the shock can cause serious harm to in-lamb ewes and new-born lambs, which become separated from their mothers. Other dogs kill poultry and may injure cattle. Puppies tend to run after sheep and poultry should be trained, at an early age, not to do so.

- **Storm-related phobias:** fear of the loud noise of thunder and slink under a bed.

  Treatment: Progressive desensitization. The dog is made to lie quietly near the owner and a very good quality tape recording of thunder is played to the dog. At first the volume is kept very low. If the dog shows no signs of apprehension it is reward with food and the volume is raised slightly. The daily session is short only 10 min. The same procedure can be used to treat other noise-related phobias as street noise, gunfire and riding cars and also non-sound related fear.

- **Biting human beings:** although dogs do not normally bite people, nervous individuals may be bitten through fear.

Thank you very much